

**Studies of temperature change in the
mesosphere using OH–emission:
Tools for monitoring climatic change.**

**Cand. Scient. Thesis
by**

Endawoke Yizengaw

May 1998



University of Tromsø
Faculty of Science
Department of Physics

TABLE OF CONTENT

Table of Content	i
Summary	iii
Acknowledgment	iv
1. INTRODUCTION	1
2. PROPERTIES OF OH EMISSION	10
2.1. Rotation and vibration of OH molecules	13
2.2. Energy level	14
2.3. Population of rotational level	16
2.4. Intensities of OH emission	18
3. THEORETICAL OH SPECTRA	19
3.1. Temperature effect of OH emission	20
3.2. The shape of OH spectral line	21
3.2.1. Doppler broadening	23
4. TOTAL ELECTRON CONTENT (TEC)	27
4.1. Diffraction grating theory	28
4.1.1. General grating properties	28
4.1.1.1. Angular dispersion	30
4.1.1.2. Linear dispersion	30
4.1.1.3. Wavelength and order	31
4.2. Ebert-Fastie Spectrometer	32
4.3. Aberration	34
4.4. Spectrometer parameter	34
4.4.1. Efficiency	35
4.4.1.1. Blazed grating	35
4.5. System characteristics	36
4.5.1. F-number of Ebert-Fastie Spectrometer	36
4.5.2. Instrument band pass	37

T A B L E o f C O N T E N T

5.	OH MEASURED SPECTRA	40
5.1.	Instrument set-up	40
5.2.	Measured spectrum	43
5.3.	Intensity of measured OH emission	47
5.3.1.	Background	49
6.	ROTATIONAL TEMPERATURE ANALYSIS	51
6.1.	Temperature extracting method	52
6.1.1.	Least-Square fitting methods	54
6.2.	Temperature variation	55
7.	DISCUSSION	60
7.1.	Mesospheric dynamics and its effect	60
7.1.1.	OH* emission diurnal variation	62
7.1.2.	Temperature variation	64
7.1.2.1.	Diurnal temperature variation	65
7.1.3.	Climatic change	67
7.2.	The effects of environmental human activity on mesospheric temperature change	68
7.3.	Turbulence and negative ions in mesosphere	69
7.4.	Estimating the wind velocity involved in the tide	70
8.	CONCLUSION	72
	Appendices	74
	Bibliography	90

Studies of temperature change in the mesosphere using OH-emission: Tools for monitoring climatic change.

Cand. Scient. Thesis

by

Endawoke Yizengaw

May 1998

Abstract: - Atmospheric temperatures from the mesosphere are deduced from spectrometer measurements of hydroxyl bands night airglow emission. Studying the temperature change in the mesosphere is a key tool for monitoring the climatic change and for prediction of the global warming. The OH (6-2) bands have been observed at a site close to Skibotn in northern Norway (69.5 °N, 20.3 °E, geographic) during January-February 1997 using Ebert-Fastie spectrometer. The background theory and instrumental set-up are described in detail, followed by a critique of sources of error. This part of the thesis provides complete background information to enable future users of the instrument to obtain temperature data quickly and easily. Necessary software is included in the appendices. An average temperature of 190.3 ± 7 K for the January campaign is obtained from a selected data set. Variations in rotational temperature at about 87 km on time scales from tens of minutes to days show similar pattern as MSISE90 at the same altitude and latitude region. The semidiurnal tide with average peak to peak amplitude of ~ 8 K is obtained. The average horizontal tidal velocity amplitude deduced from the amplitude of the semidiurnal temperature variation is 17 m s^{-1} . The effects of gravity wave propagation in the mesosphere on the rotational temperature are also discussed. Finally, strategies combining OH and incoherent scatter observations to investigate ion chemistry and support turbulence measurements are presented.

Acknowledgements

This thesis has been made possible through encouragement from my teachers and colleagues. In particular I would like to express my gratitude to my supervisor Dr. Chris Hall starting from the beginning to the very end of my thesis. Had it not been for his good outlook for those who seek knowledge, this work would never have been a reality. I am grateful to a material support and constructive discussion with my co-supervisor Prof. Asgeir Brekke. I greatly acknowledge the financial support of my work provided by University of Tromsø, department of Physics incorporation with the Norwegian State Educational Loan Fund.

The possibility of participating in the course given at UNIS, Svalbard was undoubtedly an important source of inspiration for my thesis work. I have benefited an invaluable experience gained from my five months stay at Svalbard.

I would also like to thank Prof. Ove Harang who has kindly offered me data from Skibotn, which was the key tool for my thesis work. I also appreciate all help he has given me.

At the very end of my work Truls Hansen (Associate Prof.) went through my thesis and come up with many useful suggestions about the finishing of my work. Generally, I highly appreciate all the department staffs for their collaboration, help and kindness they have been showing to their students.

My special thanks go to my fellow student Getachew Abebe for his continued help and company during my two-year stay in Tromsø, Norway. I want to thank Ethiopian National Meteorological Services (ENMS) for providing me data from Addis Ababa, which helps me relate my work to its applicability in my home country, Ethiopia. Last, certainly not least, I wish to express my gratitude to my parents and my wife for their invaluable help during my work. Without their support this work would not have been organised like this. Thanks also to my Ethiopian fellow students for their support and company during my stay.